

Joint Spectral Characterization of a Bright, Tunable, Continuous Wave Photon-Pair Source at Telecom Wavelengths

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Abstract We demonstrate a tunable fiber-coupled photon-pair source spanning $\lambda = 1490 - 1630$ nm with high internal spectral brightness ($\sim 4.6 \cdot 10^7$ pairs/s/mW/nm). Using pulse carving, we measure the joint spectral intensity and achieve tunable spectral correlations via narrow-band filtering, enabling correlation control for quantum networking applications. ©2026 The Author(s)

Introduction

Photon-pair sources are essential in quantum information applications, such as quantum networking [1, 2], optical quantum computing [3], and quantum key distribution (QKD) [4–6]. The requirements these sources must meet vary by application, such as the photon-pair generation rate and spectral characteristics. The desired spectral correlations between the signal and idler photons can be negative [7], positive [8], or uncorrelated [9]. In addition, for long-distance networking and QKD over deployed fibers, the photon pairs must be generated at telecom wavelengths to minimize losses and must be coupled into a fiber.

In this work, we investigate a fiber-coupled broadband source of type-0 spontaneous parametric down conversion (SPDC) photon pairs, tunable over $\lambda = 1490 - 1630$ nm. The source offers broadband emission at telecom wavelengths, enabling pair distribution over many channels using dense wavelength-division multiplexing [10]. In addition, we show that narrow-band filtering of the source removes the spectral correlations between the photon pair, which is critical for the indistinguishability of single photons, allowing quantum interference between independent sources.

To characterize these spectral correlations, we measure the joint spectral intensity of the photon pairs with a fiber-based single-photon spectrograph, based on the work in [11]. The primary

difference in our setup is that the spectrograph can be used with emission from continuous wave (CW) sources, which allows the spectral characterization of photon pairs generated using more affordable, free-running laser diodes. To achieve this, we employ a Mach-Zehnder interferometer to carve pulses out of the CW emission, making time-resolved coincidence measurements possible. To the best of our knowledge, the use of pulse-carving in this context has not been previously reported.

Correlated Photon-Pair Source

A schematic of the correlated photon source is shown in Figure 1 (blue box). The source consists of a fiber-coupled PPLN crystal (Covesion) with type-0 phase matching, pumped by a 780 nm CW laser (Toptica DL Pro). The phase-matching conditions of the SPDC process depend on the temperature of the crystal, allowing coarse tuning of the central wavelengths of the signal and idler photons while keeping the pump wavelength constant. The crystal temperature is controlled using an integrated heating element, regulated by a temperature controller (TC in Figure 1). The generated photon pairs are separated from the pump light in free space with two 1500 nm long-pass filters (Thorlabs FELH1500) and coupled into a single-mode fiber.

The spectral distribution of the source is measured with an optical spectrum analyzer

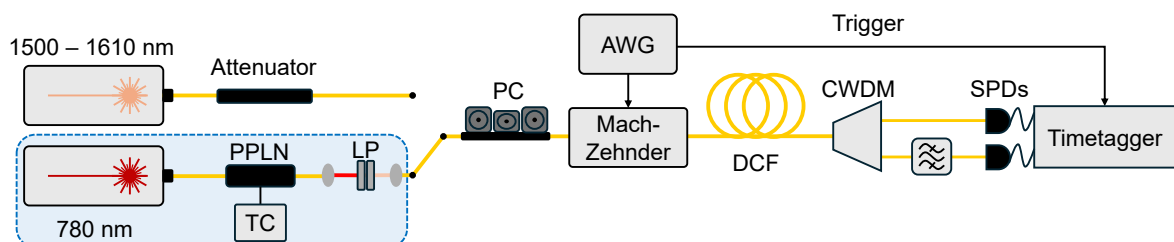


Fig. 1: Schematic of the correlated photon-pair source (blue) and fiber-based single-photon spectrograph, see text. The attenuated tunable laser is used to calibrate the time-to-wavelength conversion.

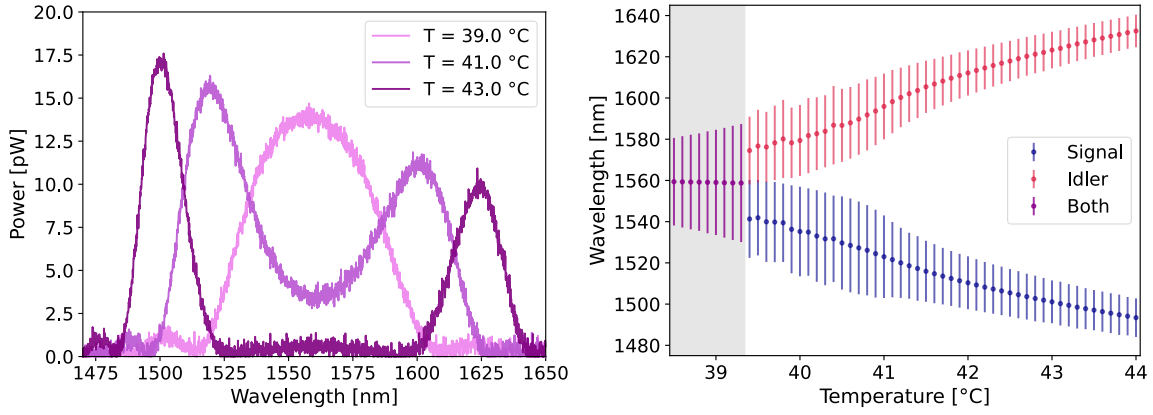


Fig. 2: Temperature dependency of the spectral distribution of the SPDC emission. **Left:** The combined spectral distribution of the signal and idler emission at a crystal temperature of 39.0 °C, 41.0 °C and 43.0 °C. **Right:** At temperatures below 39.4 °C (shaded region), the signal (blue) and idler (red) spectra overlap too much to distinguish two separate peaks. The vertical lines show the FWHM of the emission, which becomes narrower as the temperature increases.

(Yokagawa AQ6374). The distributions of the signal and idler photons are centered around twice the pump wavelength ($\sim 1560\text{ nm}$). Figure 2 shows the dependence of the central wavelengths on crystal temperature. As expected for type-0 SPDC [12], the emission exhibits a broad spectral distribution, which narrows as the temperature increases. For the remainder of the experiments, the crystal temperature is kept constant at 39.0 °C.

Next, we characterize the spectral brightness of our source. The signal and idler photons are separated with a coarse wavelength-division multiplexer (CWDM, FS 8CH Mux Demux) with 20 nm wide channels. The signal photons are filtered with a tunable bandpass filter (EXFO XTA-50/W), set to a pass wavelength of 1550 nm, with a bandwidth full width at half maximum of $\text{FWHM}_{\lambda} = 1\text{ nm}$. The photons are detected with single photon detectors (SPDs, QuantumCTek) connected to a time tagger (Swabian Ultra). The detected coincidence rates are shown in Figure 3.

At a pump power of 2.7 μW , the SPD of the

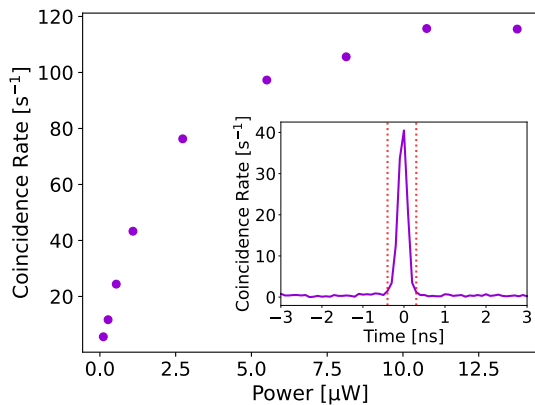


Fig. 3: Detected coincidence rates as a function of pump power. The rates approach an asymptote for pump powers above 2 μW , which is attributed to detector saturation. **Inset:** The total detected coincidence rates are obtained by integrating the coincidence rates of the relevant bins (red vertical lines).

idler path (without filter) reaches a single photon count rate exceeding 100 kHz. This approaches the saturation regime of the detector, which has a dead time of $\sim 5\text{ }\mu\text{s}$. The onset of saturation is evident from the flattening of the coincidence rates in Figure 3, which approach an asymptote. Therefore, only lower pump powers are considered for the calculation of the spectral brightness.

The detected spectral brightness is defined as [13]:

$$B_{det} = \frac{R_{cc}}{P\Delta\lambda}, \quad (1)$$

where R_{cc} is the detected coincidence rate, P the pump power and $\Delta\lambda$ the bandwidth of the tunable bandpass filter. This results in a detected spectral brightness of $B_{det} \approx 4.4 \cdot 10^4\text{ pairs/s/mW/nm}$. To estimate the internal spectral brightness of our crystal, we factor out the external losses in our system. This includes the detector efficiencies ($\eta_s = 0.253$, $\eta_i = 0.257$) and measured losses ($\alpha_s = 0.078$, $\alpha_i = 0.190$) of the signal and idler paths, resulting in an estimated internal spectral brightness of $B_{int} = B_{det}/(\alpha_s\alpha_i\eta_s\eta_i) \approx 4.6 \cdot 10^7\text{ pairs/s/mW/nm}$. This value is comparable to bright PPLN waveguide sources found in literature [14].

Joint Spectral Characterization

The experimental setup for joint spectral intensity (JSI) measurements is schematically depicted in Figure 1. To enable time-resolved coincidence measurements, a Mach-Zehnder interferometer (Fujitsu 7937EZ) driven by an arbitrary-waveform generator (AWG, Keysight M8195A) is used to carve 500 ps pulses at a rate of 12.5 MHz from the CW emission. The AWG signal triggers the timetagger to measure the time-of-arrival at the single photon detectors. A polarization controller (PC) is used to ensure alignment of the polarization with the Mach-Zehnder interferometer.

The photons are sent through a series of dispersion-compensating fiber (DCF) spools with a high absolute group-velocity dispersion (3.8 km,

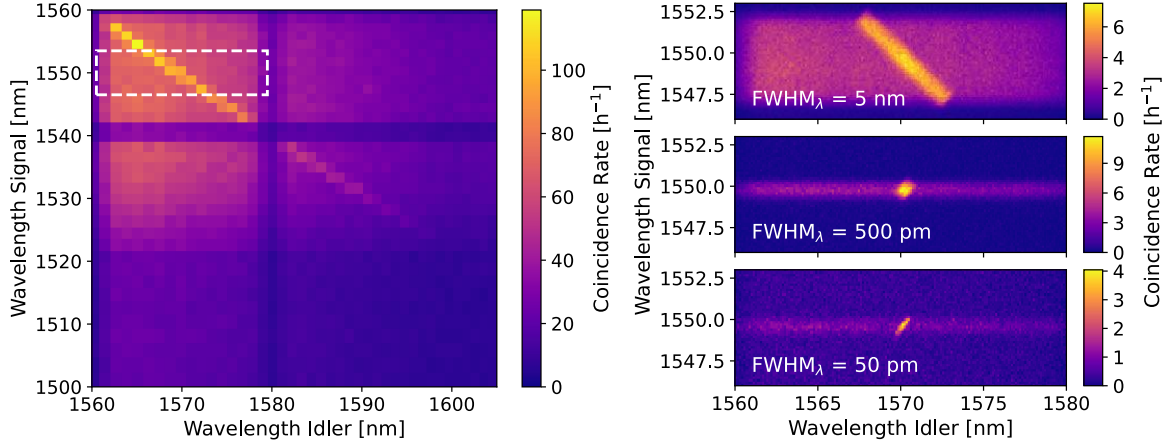


Fig. 4: Joint spectral intensity measurements without (left) and with (right) a spectral filter in the signal path. **Left:** The 'cross' of negligible coincidences at $\lambda_s = 1540$ nm and $\lambda_i = 1580$ nm originates from the boundaries between the CWDM channels. The white box indicates the regime of the filtered measurements. This measurement is taken with a bin size of 1×1 ns. **Right:** JSI measurements with spectral filter set to $\text{FWHM}_\lambda = 5$ nm (top), 500 pm (middle) and 50 pm (bottom). The bin size for these measurements is 100×100 ps.

–160 ps/nm·km), which impose a wavelength-dependent time delay. This delay is used to convert the measurement of the arrival time to the wavelength of the photons. The conversion is calibrated by probing the spectrograph with a heavily attenuated tunable laser (Santec TSL-570), sweeping the wavelength range and measuring the corresponding arrival times.

For the unfiltered JSI measurement, the full spectrum of the signal and idler photons is captured by combining the 1550, 1530 and 1510 ± 10 nm (signal) and 1570, 1590 and 1610 ± 10 nm (idler) CWDM channels with 3×1 splitters. For the filtered measurements, only the 1550 ± 10 nm (signal) and 1570 ± 10 nm (idler) channels are used. The signal photons are filtered with a tunable bandpass filter with bandwidths of $\text{FWHM}_\lambda = 5$ nm, 500 pm, and 50 pm. Joint spectral intensity plots are obtained from coincidence measurements between the SPDs, forming two-dimensional histograms of correlated arrival times.

Results & Discussion

The results of the joint spectral intensity measurements are shown in Figure 4. The unfiltered JSI has a well defined -45° slope, which indicates that our source produces photon pairs with a strong negative spectral correlation [15]. The cross-shaped feature of negligible coincidences at $\lambda_s = 1540$ nm and $\lambda_i = 1580$ nm originates from the boundaries between the CWDM channels. We attribute the majority of the noise to the finite extinction ratio of the Mach-Zehnder interferometer (≥ 20 dB), which causes leakage of generated photons with random arrival times. Coincidences between these random photons and pulse photons create a noise profile that follows the spectral distribution of the idler photons along the horizontal axis, and signal photons along the vertical axis, as is apparent in Figure 4.

The filtered JSI with $\text{FWHM}_\lambda = 5$ nm retains the same negative slope as the unfiltered JSI, showing that the signal and idler photons still have a strong negative spectral correlation in this configuration. For a filter width of $\text{FWHM}_\lambda = 500$ pm, the JSI approximates a circular shape, indicating the absence of spectral correlations between the signal and idler photons [9].

For narrower filtering with $\text{FWHM}_\lambda = 50$ pm, the shape of the JSI changes to a narrow ellipsoid with positive slope. This indicates that the signal and idler photons have a positive spectral correlation in this configuration.

Conclusion

In this work, we have characterized a tunable, fiber-coupled photon-pair source operating in the telecom wavelength regime. The source exhibits broadband emission with tunable central signal and idler wavelengths, spanning $\lambda = 1490 - 1630$ nm, and a high internal spectral brightness of $\sim 4.6 \cdot 10^7$ pairs/s/mW/nm.

We further implemented a pulse-carving technique that enables joint spectral intensity measurements of photon pairs generated from continuous-wave emission using a fiber-based single-photon spectrograph. To the best of our knowledge, the use of pulse-carving in this context has not been previously reported. These measurements show that the unfiltered source produces photon pairs with a strong negative spectral correlation, which can be tuned and completely removed through narrow-band filtering ($\text{FWHM}_\lambda \leq 500$ pm).

This control over spectral correlations is essential for achieving single-photon indistinguishability, enabling quantum interference between independent sources. These properties make the source suitable for various implementations in quantum networking and quantum key distribution.

Acknowledgements

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